



IMPRESSIVE BOMBLOAD

RAF armors check 250-lb (115-kg) bombs before they are loaded into a Stirling. Despite such careful scrutiny, about one in seven bombs dropped on Germany failed to explode.

had fewer than 500 operational front-line bombers, this involved scraping the bottom of several barrels to make up numbers, including using crews that were still in training. But at least one of the raids, on Cologne, caused considerable devastation, and a pattern was set for staging concentrated attacks by large-scale forces against a single city.

Bomber crews

At the start of the war RAF bomber crews were often poorly trained.

Apart from the pilots, they consisted mostly of young men who had

joined the air force to learn a trade and then signed up to fly for the extra pay. As training improved, the specialized skills of the navigator, radio operator, bomb-aimer, and gunners

complemented those of the pilot. At the end of training, men were not usually allotted to a crew, but teamed up through a process of self-selection – they were simply put together and told to group themselves into crews. Once formed, a crew traditionally stuck together on the ground as in the air. Lancaster pilot Jack Currie observed that for “a gunner to be in company with the pilot of another crew more than once or twice would be thought unnatural or disloyal.” On a mission, all the members of the crew depended upon one another for survival; they lived or died together. One British flier observed that “friendships thus forged had a depth and unique quality that never existed with friendships before, and for me never after.”

RADARS

THE NIGHT FIGHTING over Germany during World War II accelerated the development of compact airborne radar sets. The Luftwaffe, for example, equipped Bf 110 and Ju 88 night fighters with Lichtenstein interception radar, which greatly improved the aircraft's effectiveness. Manned by the radio operator, it had a range of 2 miles (3km), and was used when the aircraft were closing in on targets. British bombers were equipped with H2S radar sets (below) to help them find their targets. Various radar-jamming devices were also deployed by both sides. Radar detectors enabled the Germans to locate enemy bombers by their H2S emissions, while similar equipment warned bomber crews of the approach of radar-guided night fighters.

PPI (Plan Position Indicator, map display)

Range control (in miles)

Height scale (in feet)

Slant/ground range-calculation scale

Height tube

H2S RADAR SYSTEM

The RAF's H2S airborne radar helped night bombers identify targets on the ground. It was also used by the USAAF on daylight raids when there was heavy cloud.

